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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester (CBCS) Degree Examinations, NOV. - DEC. 2025**25PHY102– QUANTUM COMPUTING AND MODERN PHYSICS**

(For AID, AIM, CYB, CCE, CSE, CBS, ISE, RAI)

Duration: 3 Hours]

[Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from each Unit.

List of Constants:Velocity of light, $c = 3 \times 10^8 \text{ ms}^{-1}$,Planck's constant, $h = 6.63 \times 10^{-34} \text{ Js}$,Electron mass, $m = 9.11 \times 10^{-31} \text{ kg}$,Electron charge, $e = 1.6 \times 10^{-19} \text{ C}$,Permeability of the free space $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$,Permittivity of vacuum, $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$,Boltzmann constant, $k = 1.38 \times 10^{-23} \text{ J/K}$,Avogadro number, $N_A = 6.023 \times 10^{26} / \text{kg mole}$,Mass of Proton/Neutron = $1.678 \times 10^{-27} \text{ kg}$ **PART - A: MULTIPLE CHOICE QUESTIONS****20 Marks**

- The de Broglie wavelength of a charged particle of charge 'e' accelerated through a potential of 'V' volts is:
A) $\lambda = \sqrt{2meV} / h$ B) $\lambda = h / meV$
C) $\lambda = h / \sqrt{2meV}$ D) $\lambda = h/2meV$
- The group velocity of matter waves associated with a moving particle is:
A) The same as phase velocity B) Less than the particle velocity
C) More than the particle velocity D) Equal to the particle velocity
- The non-existence of electron inside nucleus is explained by
A) Relativity B) Heisenberg uncertainty principle
C) Pauli exclusion principle D) Coulomb force
- According to Max Born's interpretation, $|\psi|^2$ represents
A) probability density B) energy density
C) particle density D) charge density
- Which component of a computer is primarily affected by Moore's Law?
A) Central Processing Unit (CPU) B) Random Access Memory (RAM)
C) Hard Disk Drive (HDD) D) Graphics Processing Unit (GPU)
- Which property allows qubits to represent multiple states simultaneously in quantum computing?
A) Entanglement B) Interference
C) Superposition D) Tunnelling

7. If $\langle u|v \rangle = 0$, then u and v are said to be:
- A) Linearly dependent B) Orthogonal
C) Orthonormal D) Equal
8. This quantum gate acts on a single qubit and would most be similar to a traditional NOT gate
- A) CNOT B) X-Gate
C) Hadamard D) Deutsch Gate
9. Mobility of electron is
- A) Average flow of electrons per unit electric field. B) Average applied field per unit drift velocity.
C) Average drift velocity per unit electric field. D) Reciprocal of conductivity per unit charge.
10. Quantum free electron theory assumes electrons obey:
- A) Bose-Einstein statistics B) Maxwell-Boltzmann statistics
C) Fermi-Dirac statistics D) Classical statistics
11. The magnetic susceptibility (χ) in superconductor is
- A) positive B) negative
C) zero D) infinity
12. The electron pairs in a superconductor are _____
- A) Bosons B) Leptons
C) Hydrons D) Fermions
13. When a pentavalent impurity is added to a pure semiconductor, it becomes
- A) An insulator B) An intrinsic semiconductor
C) p-type semiconductor D) n-type semiconductor
14. The donor energy level in an n-type semiconductor is located:
- A) In the middle of the forbidden energy gap B) Just below the conduction band
C) Just above the valence band D) Far above the conduction band
15. What happens to the Hall voltage in a conductor if the magnetic field direction is reversed?
- A) The Hall voltage becomes zero. B) The Hall voltage changes direction but remains the same magnitude.
C) The Hall voltage doubles in magnitude. D) The Hall voltage remains unchanged.
16. The primary function of a solar cell is
- A) To store solar energy in chemical form B) To reflect sunlight to solar heaters
C) To convert sunlight directly into electrical energy D) To absorb and retain heat from sunlight
17. The purpose of the optical cavity in a laser is to:
- A) Convert electrical energy into light B) Amplify the light through multiple passes between mirrors
C) Filter out unwanted frequencies of light D) Focus the light into a beam
18. A semiconductor laser has a peak emission radiation of wavelength $1.24 \mu\text{m}$. Its energy gap value in eV is
- A) 1.4 eV B) 1.6 eV
C) 1 eV D) 1.8 eV
19. Which of the following has more distortion?
- A) Single step-index fiber B) Graded index fiber
C) Multimode step-index fiber D) Glass fiber
20. Which fiber is preferred for long distance communication?
- A) Step index single mode fiber B) Graded index multimode fiber
C) Step index multimode fiber D) Both single mode and multimode fibers

PART - B: DESCRIPTIVE ANSWER QUESTIONS

80 Marks

Unit – I

	Marks	BT*	CO*	PO*
1. a) Derive Schrodinger's time-independent one-dimensional wave equation for a particle of mass m with energy E .	7	2	1	1,2
b) Explain the concepts of group velocity and phase velocity and derive the relationship between them.	5	2	1	1,2
c) Compare the energy of a photon with that of a neutron when both are associated with a wave length of $1\text{\AA}$, given that the mass of neutron is $1.68 \times 10^{-27} \text{ Kg}$.	4	3	1	1,2
2. a) Assuming the solution of Schrodinger's wave equation for a particle in an infinitely deep potential well, discuss the wave function, probability density and energy eigen values for first three energy levels.	7	2	1	1,2
b) Explain Heisenberg's uncertainty principle. Show that an electron does not exist inside the nucleus.	5	2	1	1,2
c) The uncertainty in the measurement of time spent by Iridium -191 nuclei in the excited state is found to be 1.4×10^{-10} seconds. Estimate the uncertainty that results in its energy in the excited state.	4	3	1	1,2

Unit – II

3. a) What is Bloch sphere? Explain the representation of single qubit and any three of its superposition states in Bloch sphere.	8	2	2	1,2
b) Write any four differences between classical and quantum computing	4	2	2	1,2
c) Given $ \psi\rangle = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}$ and $ \phi\rangle = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$ prove that $\langle\psi \phi\rangle = \langle\phi \psi\rangle^*$	4	3	2	1,2
4. a) Explain Hadamard gate and its operation on qubits with truth table.	8	2	2	1,2
b) Explain the normalization and orthogonality properties of quantum states. Verify if $ \psi\rangle = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $ \phi\rangle = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ are orthogonal.	4	3	2	1,2
c) If $ \psi\rangle = \begin{bmatrix} 3+i \\ 4-i \end{bmatrix}$ and $ \phi\rangle = \begin{bmatrix} 3i \\ 4 \end{bmatrix}$. Find the inner product of ψ with ϕ	4	3	2	1,2

Unit – III

5. a) Obtain an expression for the electrical conductivity of a metal based on the classical free electron theory.	8	2	3	1,2
b) Discuss the drawbacks of classical free electron theory.	4	2	3	1,2
c) The critical temperature and critical magnetic field at 0 K for superconducting lead are 7.2 K and 800 gauss respectively. What will be the temperature up to which lead will be in superconducting state in a magnetic field of 400 gauss?	4	3	3	1,2
6. a) Explain behavior of Type-I and Type-II superconductors as a function of strength of applied magnetic field.	7	2	3	1,2
b) Explain the Fermi factor for different temperature conditions.	5	2	3	1,2
c) Find the temperature at which there is 2% probability that an energy level 0.3 eV above the Fermi level is being occupied.	4	3	3	1,2

Unit – IV

7. a) State and explain Hall effect. Derive an expression for Hall voltage.	7	2	4	1,2
b) Differentiate direct and indirect band-gap semiconductors.	5	2	4	1,2

c)	The resistivity of germanium at 20° C is 0.5 ohm m. What will be its resistivity at 40°C if the band gap of germanium is 0.7 eV?	4	3	4	1,2
8. a)	What are intrinsic semiconductors? Obtain an expression for electrical conductivity of an intrinsic semiconductor.	7	2	4	1,2
b)	Explain the construction and working of a photodiode.	5	2	4	1,2
c)	A sample of silicon semiconductor is doped with 10^{22} phosphorous atoms. Calculate its conductivity if mobility of electrons is $0.07 \text{ m}^2/\text{V-s}$. What is the Hall voltage if this semiconductor with a thickness of $100 \mu\text{m}$ and carrying a current of 1 mA is placed perpendicular to a magnetic field of 0.1 T?	4	3	4	1,2

Unit – V

9. a)	Describe the construction and working of a semiconductor diode laser with necessary diagrams.	7	2	5	1,2
b)	What is attenuation? Explain any two mechanisms involved in attenuation.	5	2	5	1,2
c)	The ratio of population of two energy level is 1.08×10^{-30} . Calculate the temperature, if the wavelength of the emitted photon is 696 nm.	4	3	5	1,2
10. a)	Explain the ray propagation through an optical fiber and derive expressions for the numerical aperture and acceptance angle of an optical fiber.	7	2	5	1,2
b)	Explain the terms Metastable state and Population inversion. Explain the necessity of population inversion for lasing.	5	2	5	1,2
c)	Calculate the acceptance angle and critical angle for the core-cladding interface when the core R.I. is 1.48 and fractional index change is 0.02.	4	3	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
